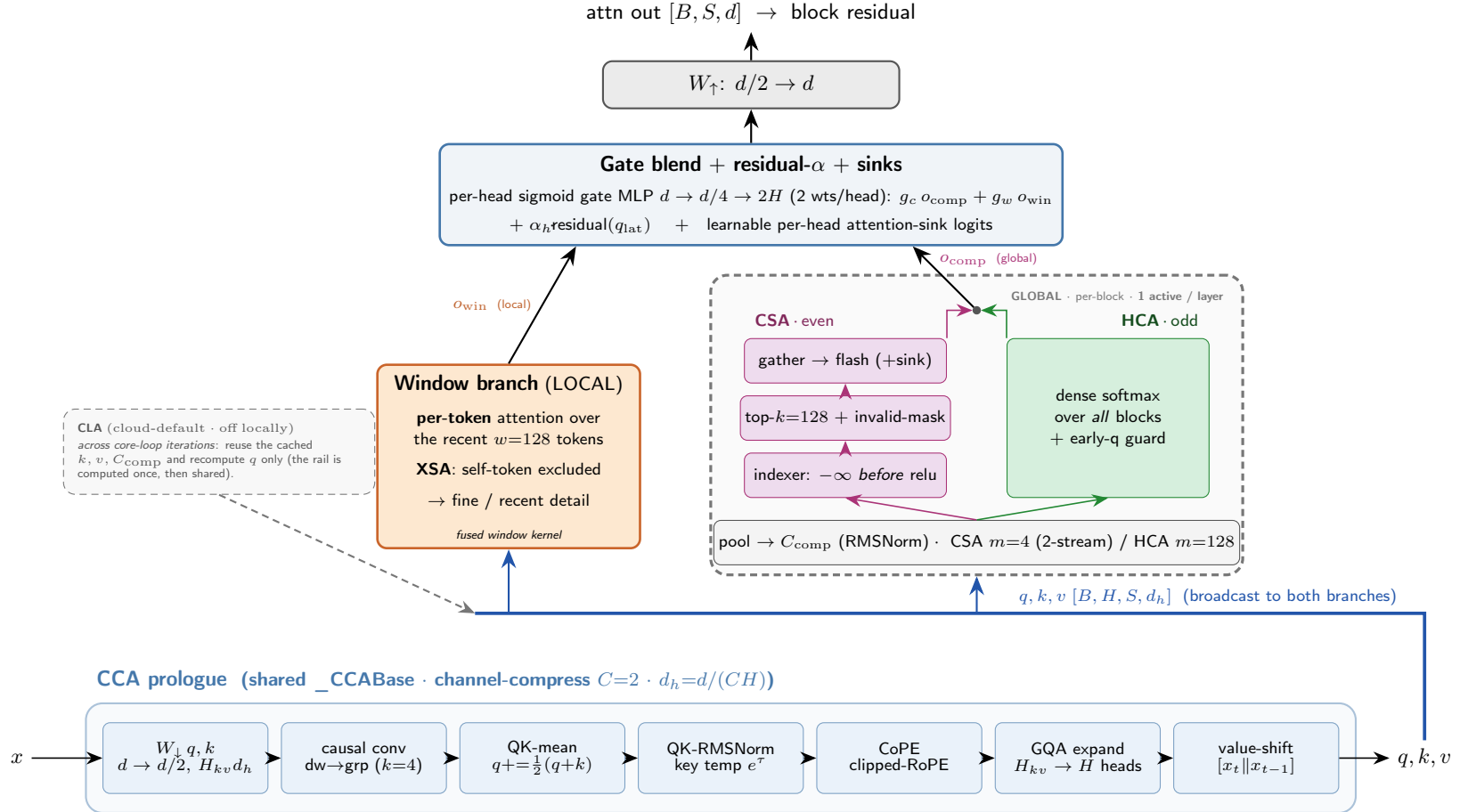


MORPH Attention — Triple-Axis Compression

a shared CCA channel-compress front-end feeds a local window || one global-compressed branch (CSA / HCA), blended by a per-head gate



Four compression axes (faithful to morph/model/attention.py): channel CCA $E \rightarrow E/2$ select CSA top-128 of $S/4$ blocks dense HCA all $S/128$ blocks local window $w=128$.

CSA/HCA alternate on the *global* layer index (even $\rightarrow$ CSA, odd $\rightarrow$ HCA), baked at `__init__` — a looped core layer keeps its variant on every iteration (*not* iteration-conditioned).

Fused Triton: CCA prologue, conv, window, CSA, HCA (each fwd/grad-exact). **Notation**: d model dim, H q-heads, H_{kv} kv-heads (GQA), channel-compress $C=2$, $d_h=d/(CH)$, latent $q=d/2$, $n_{blk}=S/m$. Color secondary — every box is text-labeled.